higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING (CAD COMPONENT)

(Second Paper)

NQF LEVEL 4

(12050674)

14 November 2019 (X-Paper)
09:00–11:00

REQUIREMENTS: **A4 drawing sheets**
Yellow cover folder (BOE 8/15)

Calculators may be used.

This question paper consists of 4 pages.

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TIME: 2 HOURS
MARKS: 75

INSTRUCTIONS AND INFORMATION

1. Read ALL the questions carefully.
2. Answer ALL the questions.
3. Number the answers according to the numbering system used in this question paper.
4. Use only ONE side of the DRAWING SHEETS.
5. Draw a 15 mm border on both sides of the DRAWING SHEETS.
6. Use your discretion where applicable.
7. Insert ALL loose DRAWING SHEETS in the cover folder – do NOT staple.
8. Work neatly.

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QUESTION 1: REINFORCED-CONCRETE DETAILING AND SPECIFICATIONS

1.1 Produce a drawing of a cantilever retaining wall using the following details:

Height of wall:	4 900 mm
Thickness of wall (top):	800 mm
Thickness of base:	1 000 mm
Length of toe:	1 500 mm
Length of tail:	4 200 mm
Concrete cover:	45 mm

- The wall extends 350 mm above the ground level
- The tail is embedded 4 550 mm below the ground and situated on the Right-hand side of the wall.
- The toe is 900 mm below the ground.
- The wall is reinforced longitudinally with Y16 @ 500 c/c and transversely with Y20s @ 300 c/c.
- The base is reinforced longitudinally with Y16 @ 500 c/c and transversely with Y20s @ 250 c/c. (30)

1.2 Redraw the culvert in FIGURE 1 using the following details:

Total length of culvert:	2 200 mm
Height of culvert:	2 200 mm
Distance to ground level:	1 200 mm
Type of culvert:	Box

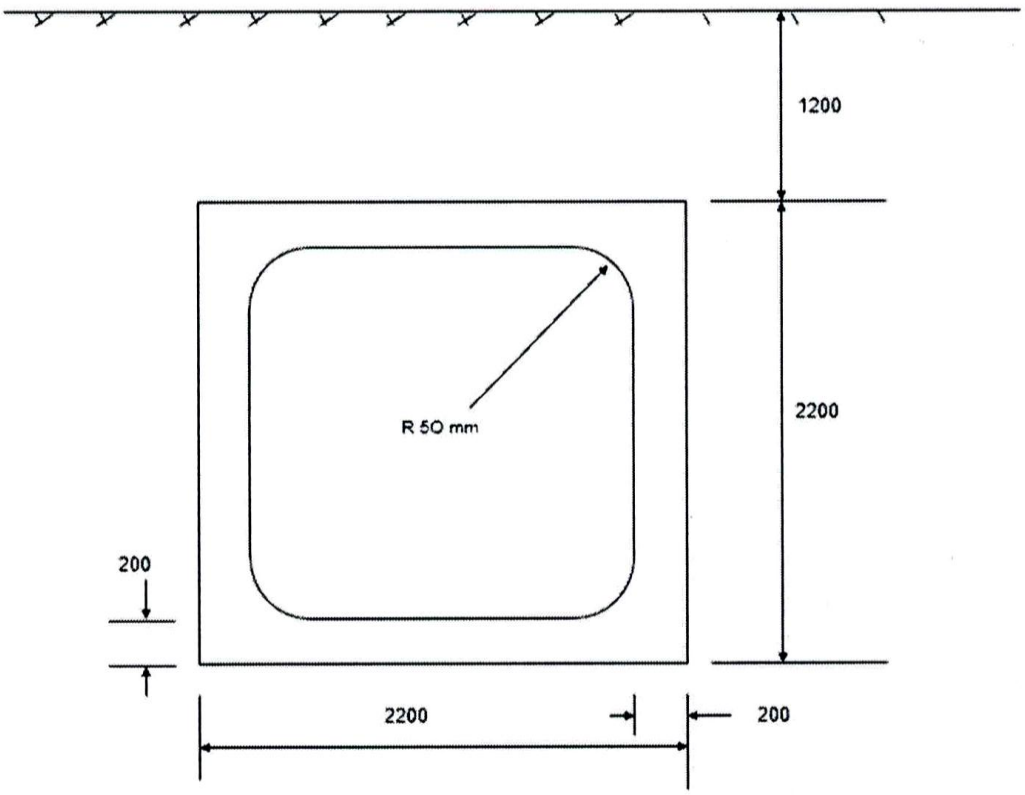


FIGURE 1 (20)

[50]

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QUESTION 2: STRUCTURAL STEEL WORK DETAILING AND SPECIFICATIONS

2.1 Draw the side view and connection details of a steel-work staircase using CAD.

Specifications:

Total height:	1 660 mm
Total length:	1 900 mm
Going:	250 mm
Rise:	200 mm
Nosing:	25 mm
Waist:	150 mm
Stringer:	250 mm × 12 mm flat
Riser:	175 mm × 8 mm flat
Slope:	34°
Tread:	15 mm

(15)

The stairs are reinforced longtitudinally with y 16 @ 300 c/c
The stairs are reinforced traversely Y16 per stair

2.2 FIGURE 2 shows a gusset plate used in a steel truss. The gusset is cut from an 8 mm steel plate.

Redraw the gusset plate using CAD. Include ALL essential dimensions.

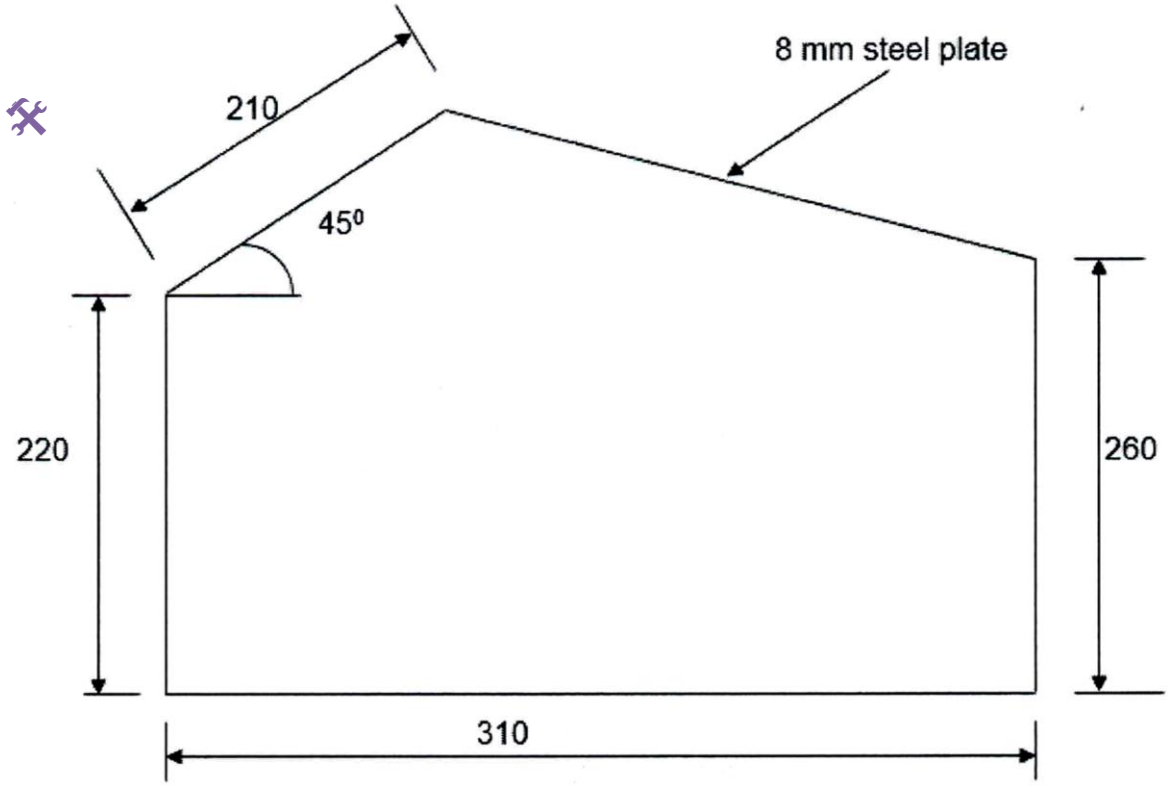


FIGURE 2

(10)
[25]

TOTAL: 75

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